An Exploration

Jack Dumanski

Executive Summary

Data Cleaning

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor

Attaching package: 'zoo'


The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
```

Residents Analysis

Question 2 with Questions 11 and 12

```
Analysis of Deviance Table

Model 1: cleft_response ~ cleft_center + program
Model 2: cleft_response ~ 1
```

Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
Q8a	Q2	0.001	TRUE	fisher	0.0033	TRUE
Q8b	Q2	0.001	TRUE	fisher	0.0033	TRUE
Q8c	Q2	5e-04	TRUE	fisher	0.0033	TRUE
Q8d	Q2	0.0045	TRUE	fisher	0.0112	TRUE
Q12	Q2	0.1134	FALSE	fisher	0.2268	FALSE
Q7b	Q2	0.6992	FALSE	fisher	0.8607	FALSE
Q7c	Q2	0.7746	FALSE	fisher	0.8607	FALSE
Q7d	Q2	0.5217	FALSE	fisher	0.8607	FALSE
Q11	Q2	0.6587	FALSE	fisher	0.8607	FALSE
Q7a	Q2	0.8701	FALSE	fisher	0.8701	FALSE

```
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        73     89.195
2        76     99.769 -3  -10.573  0.01427 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	(1)
(Intercept)	1.059
	(0.865)
cleft_centerYes	0.713
	(0.758)
programHospital-based	−1.635
	(0.623)
programUniversity	0.236
	(1.290)
Num.Obs.	77
AIC	97.2
BIC	106.6
Log.Lik.	−44.598
F	2.950
RMSE	0.45

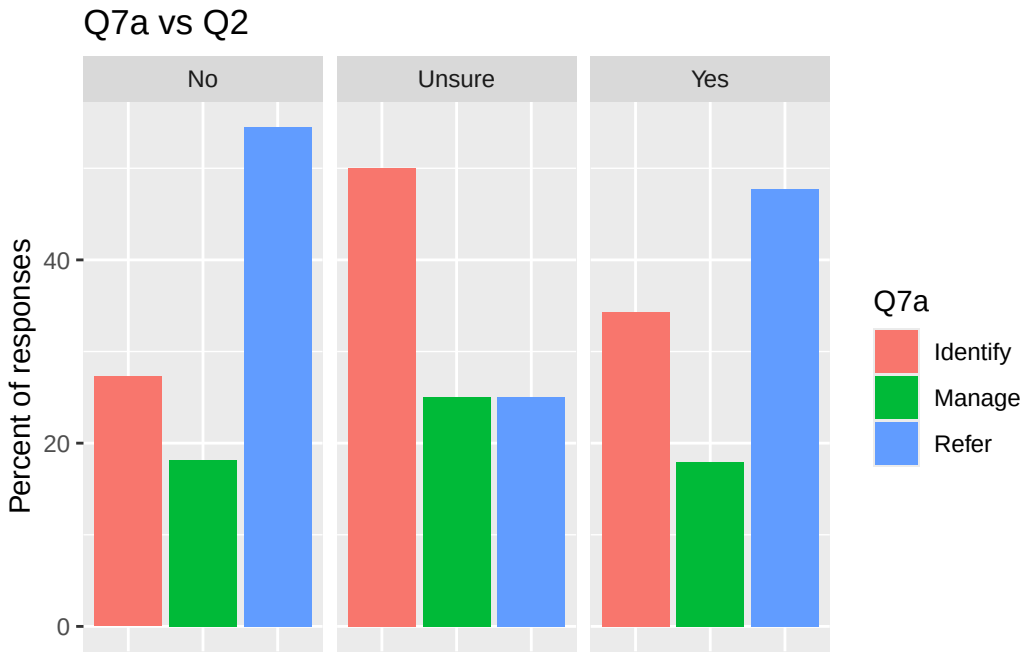
Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
Q6	role	0.0025	TRUE	fisher	0.0288	TRUE
Q8c	role	0.0015	TRUE	fisher	0.0288	TRUE
Q7b	role	0.0085	TRUE	fisher	0.0517	FALSE
Q8b	role	0.009	TRUE	fisher	0.0517	FALSE
Q8a	role	0.0115	TRUE	fisher	0.0529	FALSE
Q10a	role	0.031	TRUE	fisher	0.1188	FALSE
Q9	role	0.0375	TRUE	fisher	0.1232	FALSE
Q7d	role	0.046	TRUE	fisher	0.1322	FALSE
Q7c	role	0.0675	FALSE	fisher	0.1725	FALSE
Q13	role	0.0755	FALSE	fisher	0.1736	FALSE
Q7a	role	0.089	FALSE	fisher	0.1861	FALSE
Q5	role	0.1062	FALSE	fisher	0.2036	FALSE
Q8d	role	0.1409	FALSE	fisher	0.2493	FALSE
Q4	role	0.1769	FALSE	fisher	0.2906	FALSE
Q10e	role	0.2779	FALSE	fisher	0.4261	FALSE
Q10c	role	0.4053	FALSE	fisher	0.5826	FALSE
Q10b	role	0.4953	FALSE	fisher	0.6329	FALSE
Q12	role	0.4863	FALSE	fisher	0.6329	FALSE
Q10d	role	0.5932	FALSE	fisher	0.7181	FALSE
Q2	role	0.8376	FALSE	fisher	0.9632	FALSE
Q15	role	0.8961	FALSE	fisher	0.9814	FALSE
Q3	role	1	FALSE	fisher	1.0000	FALSE
Q14	role	1	FALSE	fisher	1.0000	FALSE

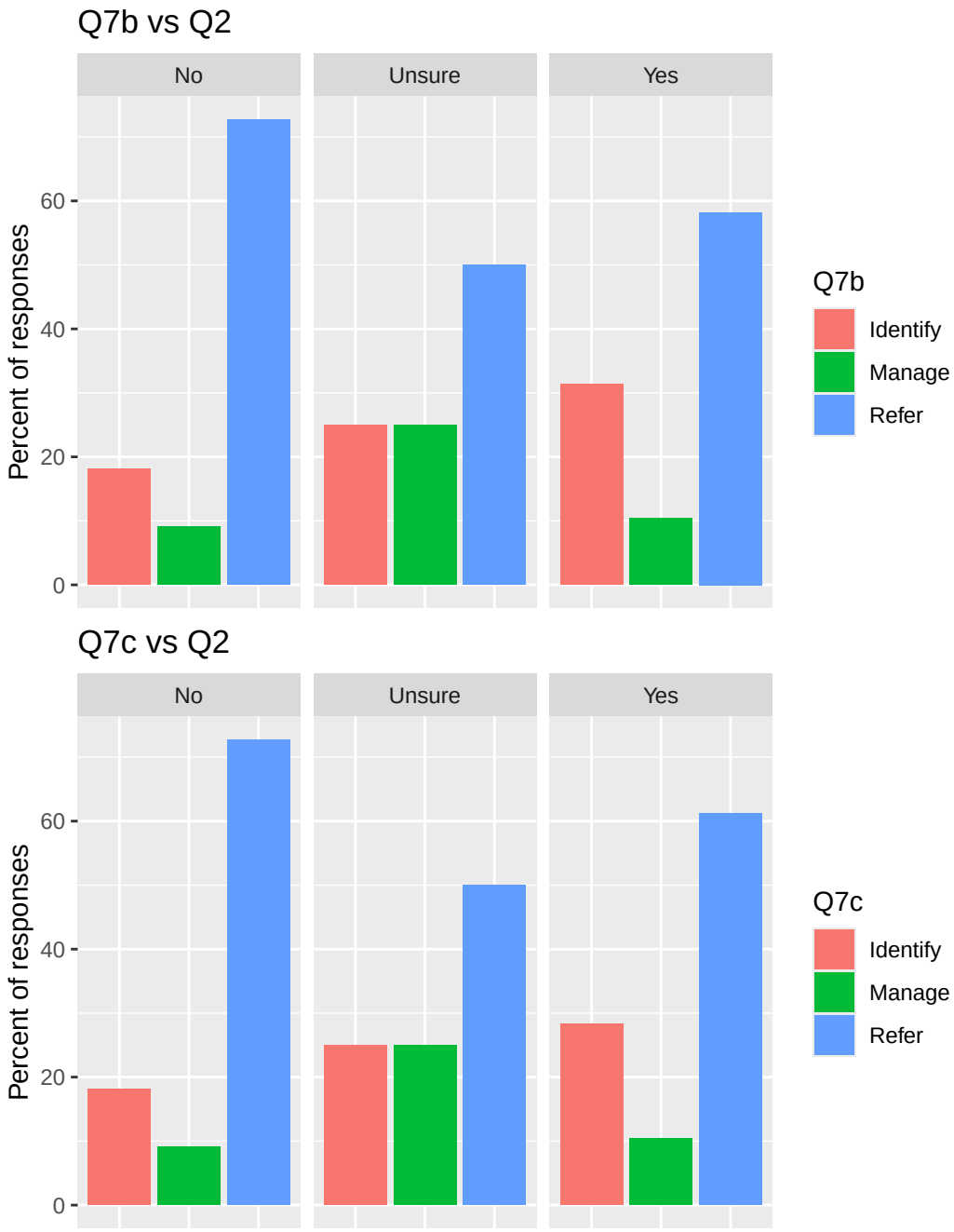
The logistic regression summary

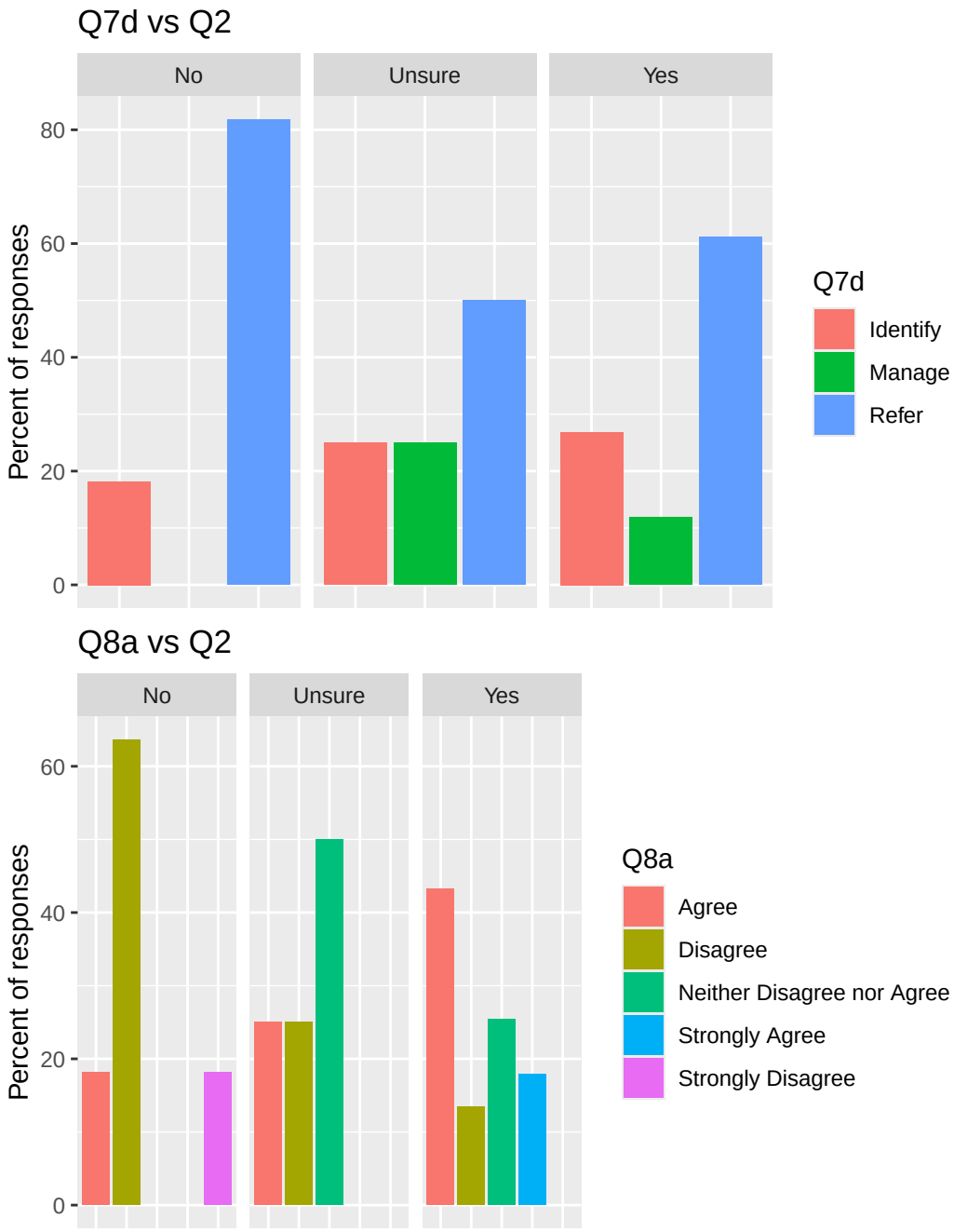
Residents & Directors Cross-Referenced

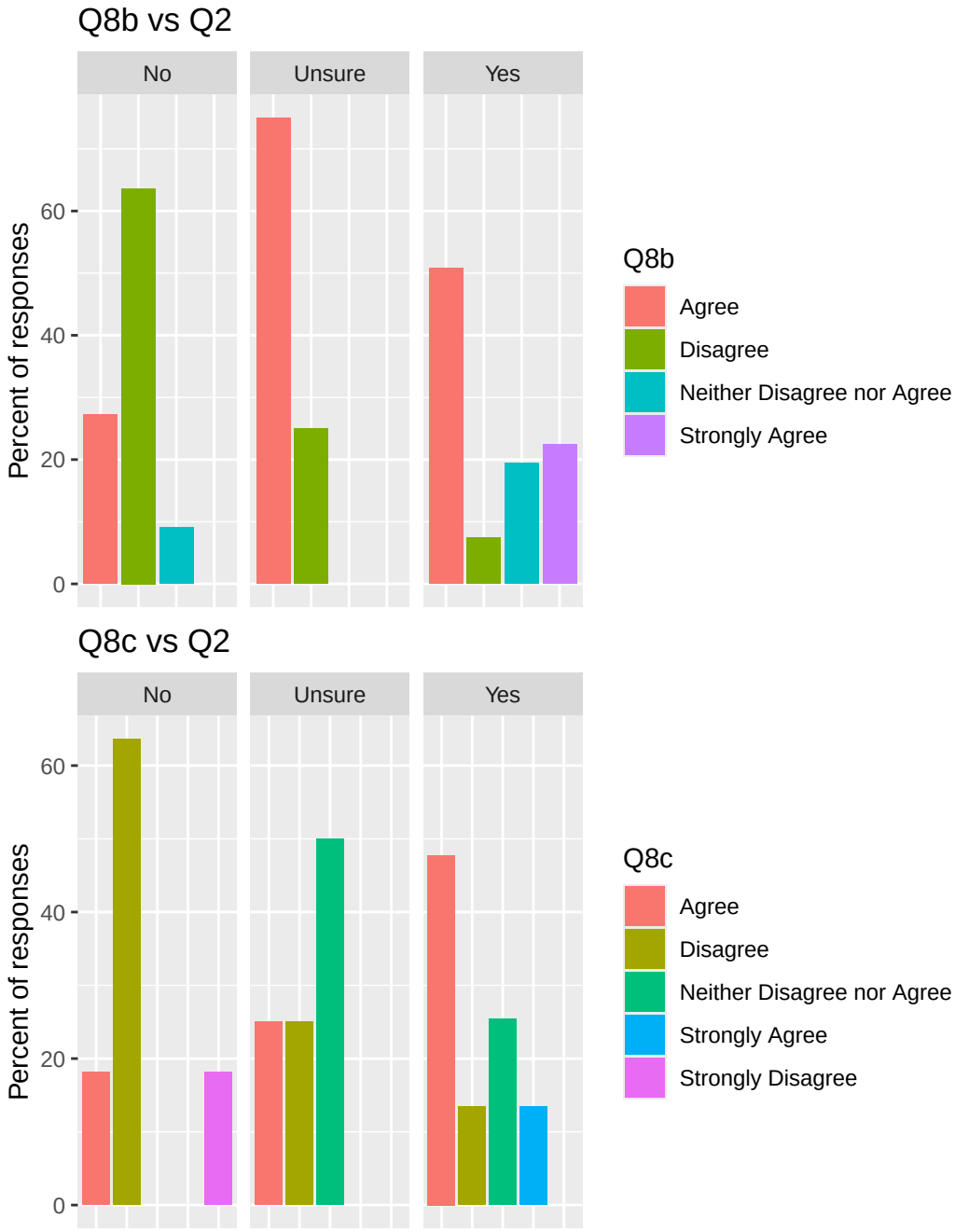
Run the tests

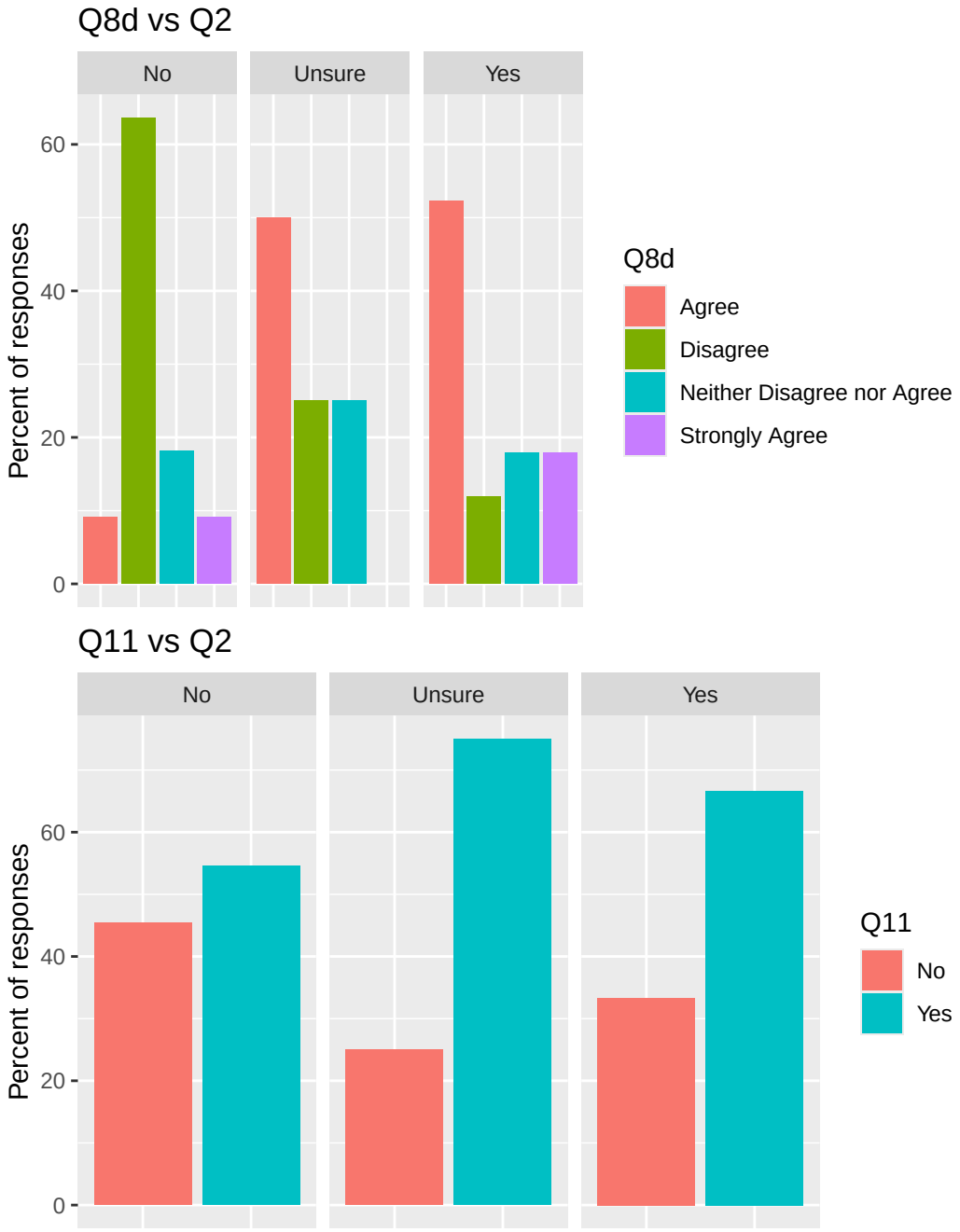
Visualization

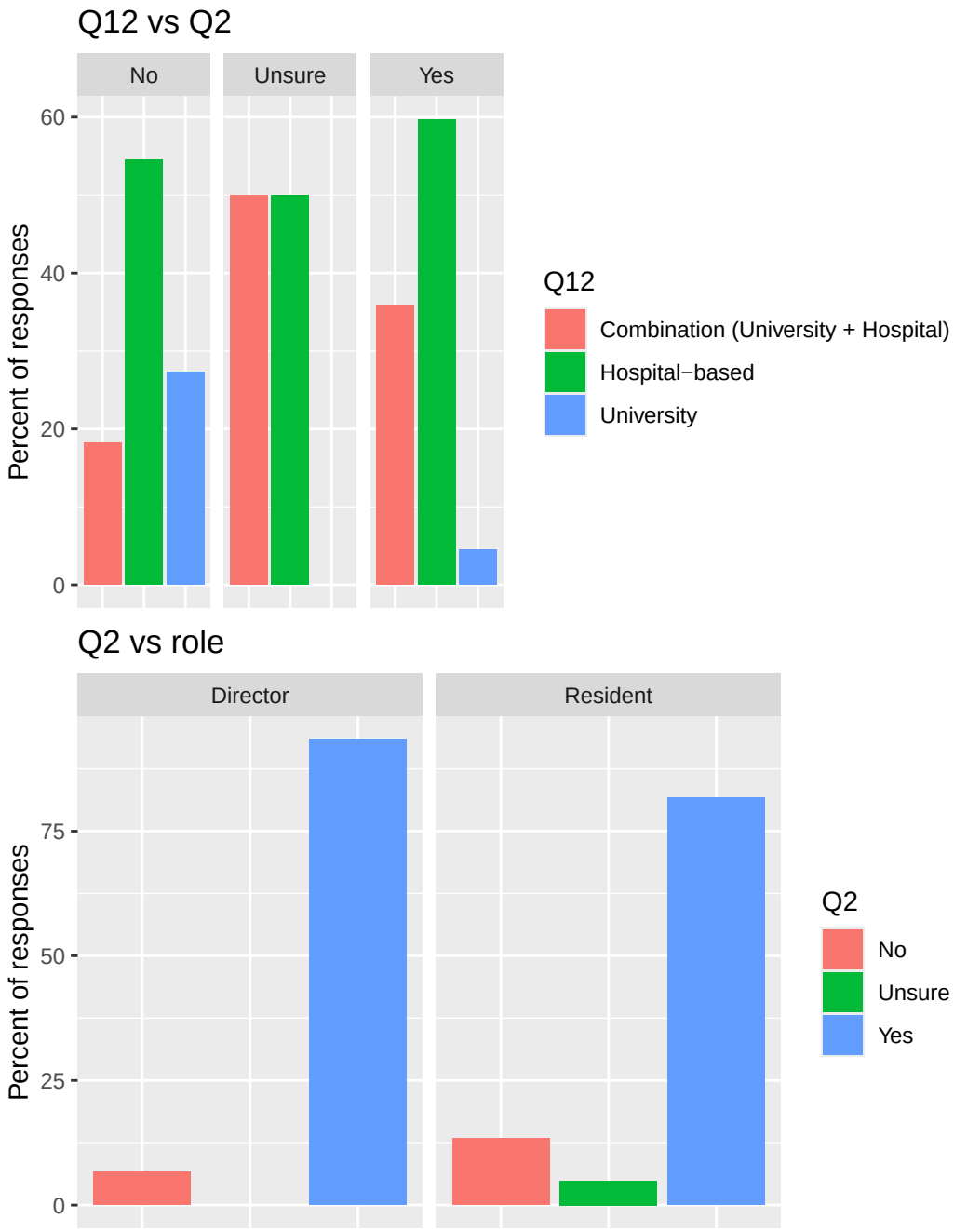


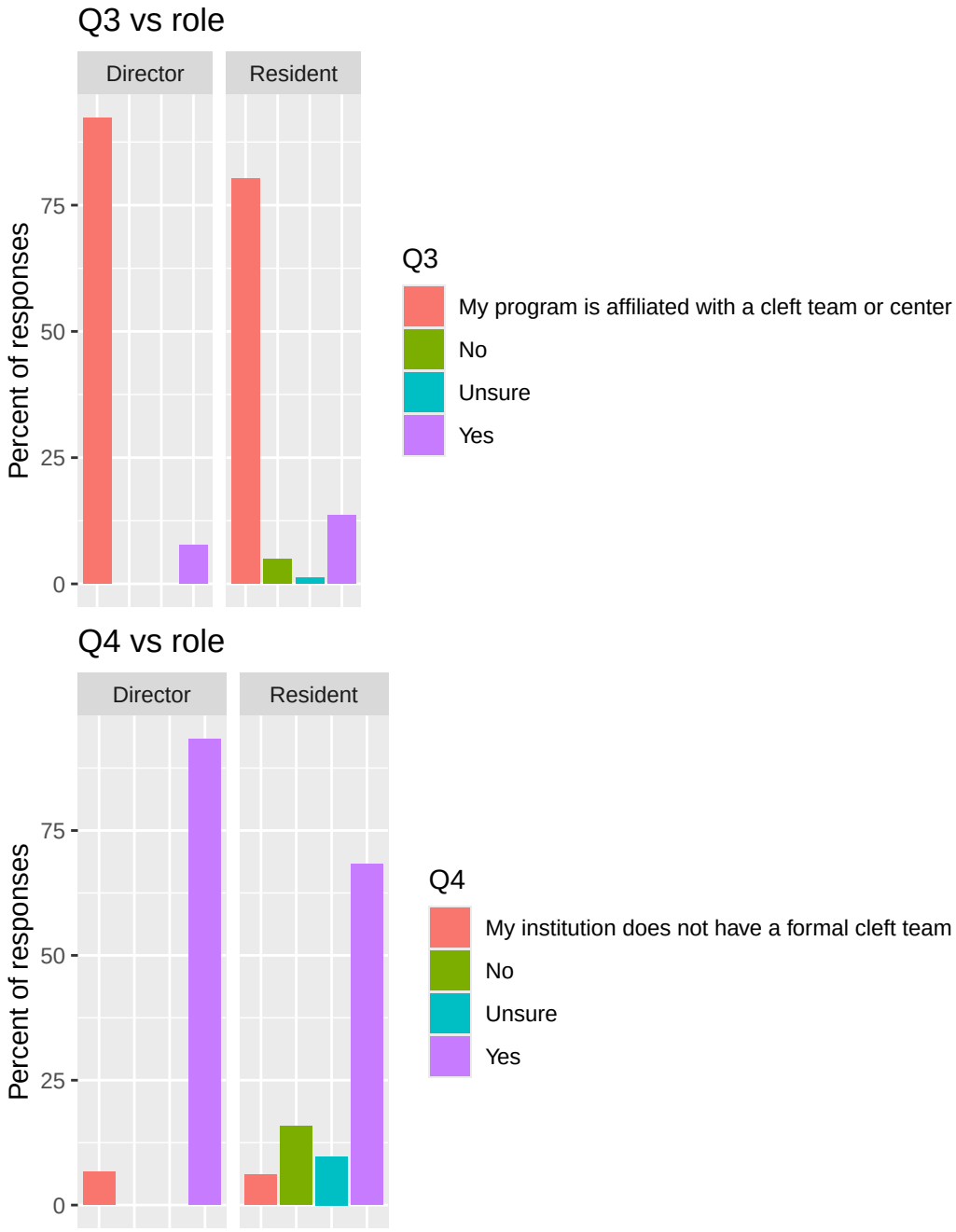

Q7b vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q7b
Identify
Manage
Refer
Q7c vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q7c
Identify
Manage
Refer

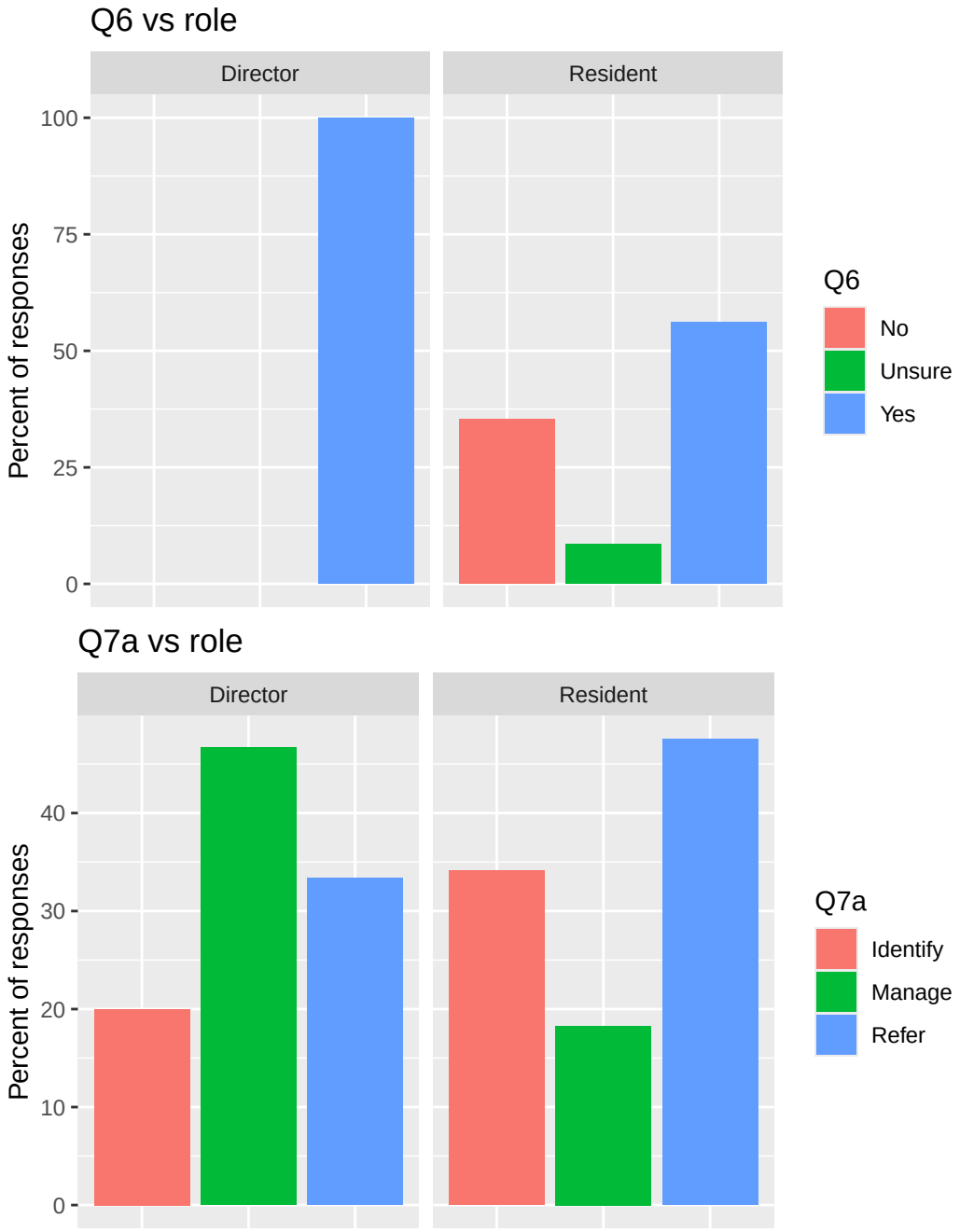

Q7d vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
80
Q7d
Identify
Manage
Refer
Q8a vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q8a
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree

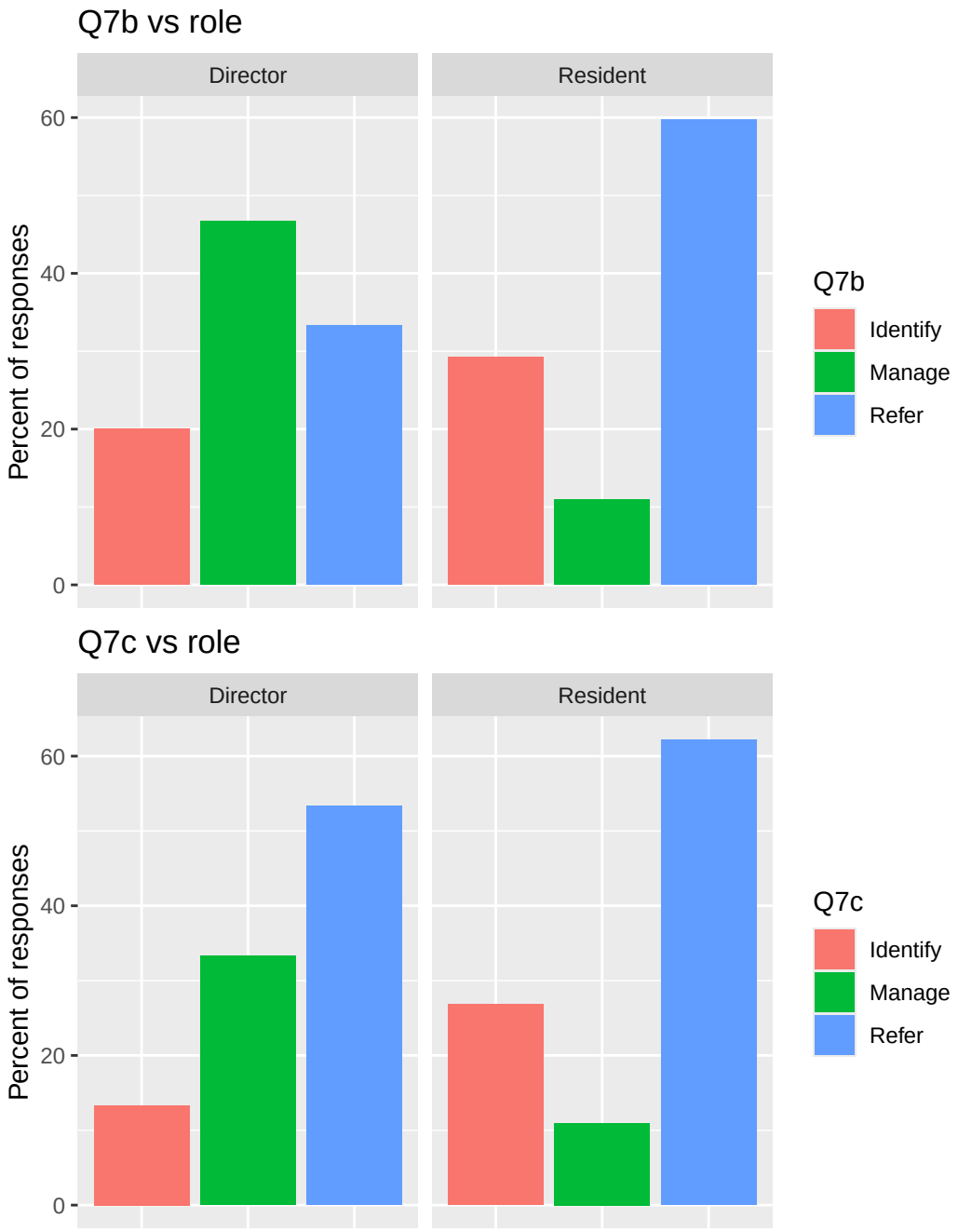

Q8b vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q8b
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Q8c vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q8c
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree

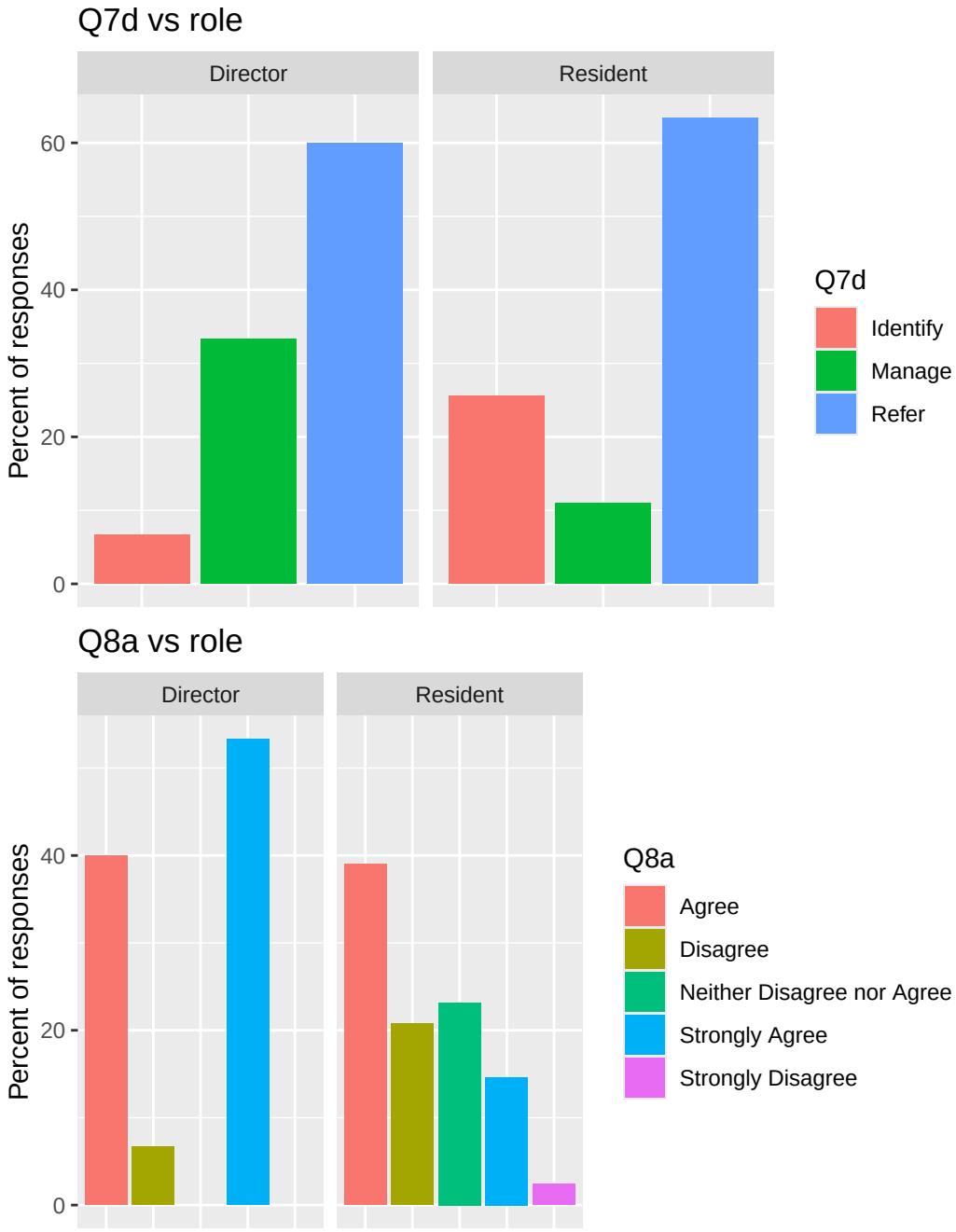

Q8d vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q8d
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Q11 vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q11
No
Yes

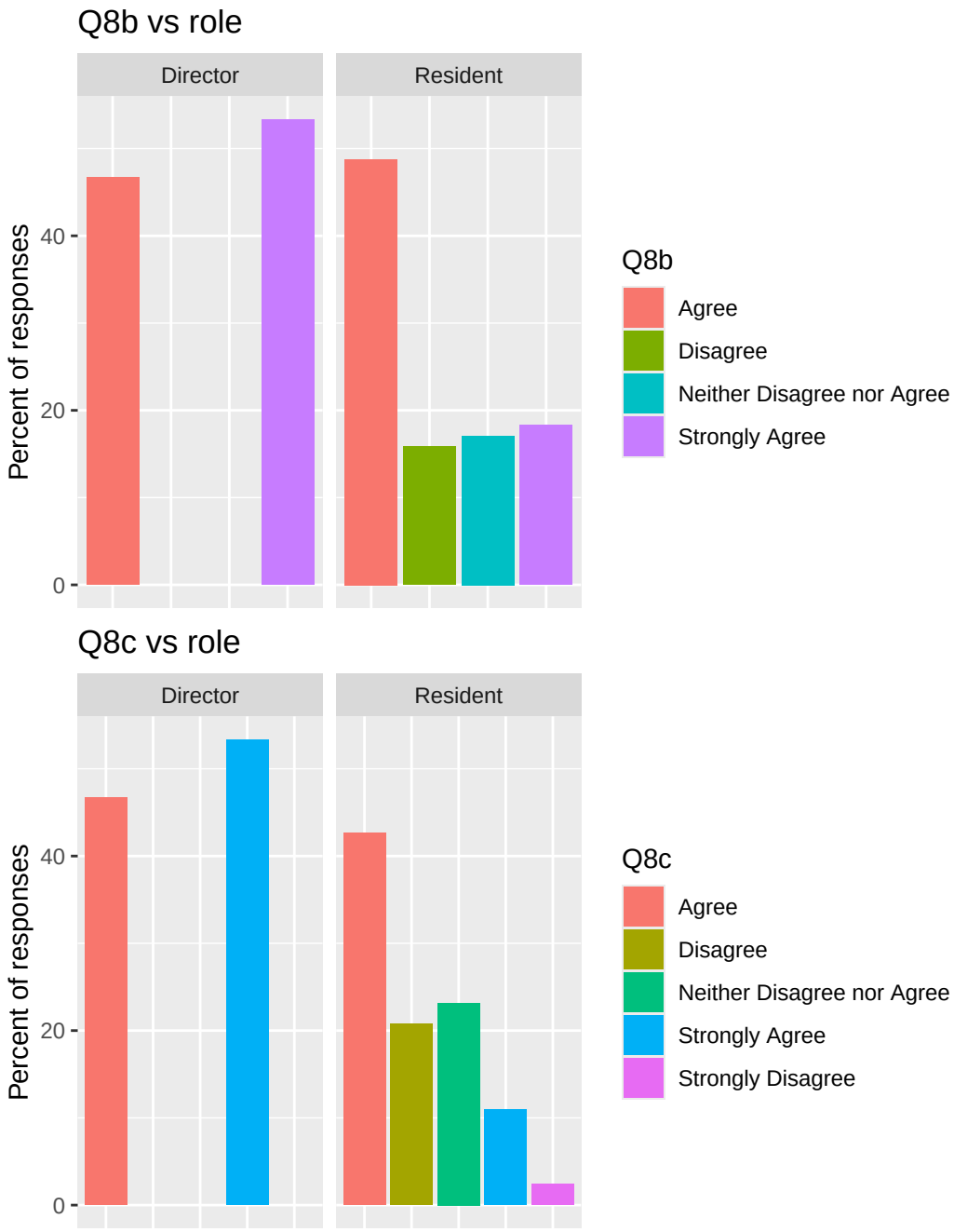

Q12 vs Q2
No
Unsure
Yes
Percent of responses
60
40
20
0
Q12
Combination (University + Hospital)
Hospital−based
University
Q2 vs role
Director
Resident
Percent of responses
75
50
25
0
Q2
No
Unsure
Yes

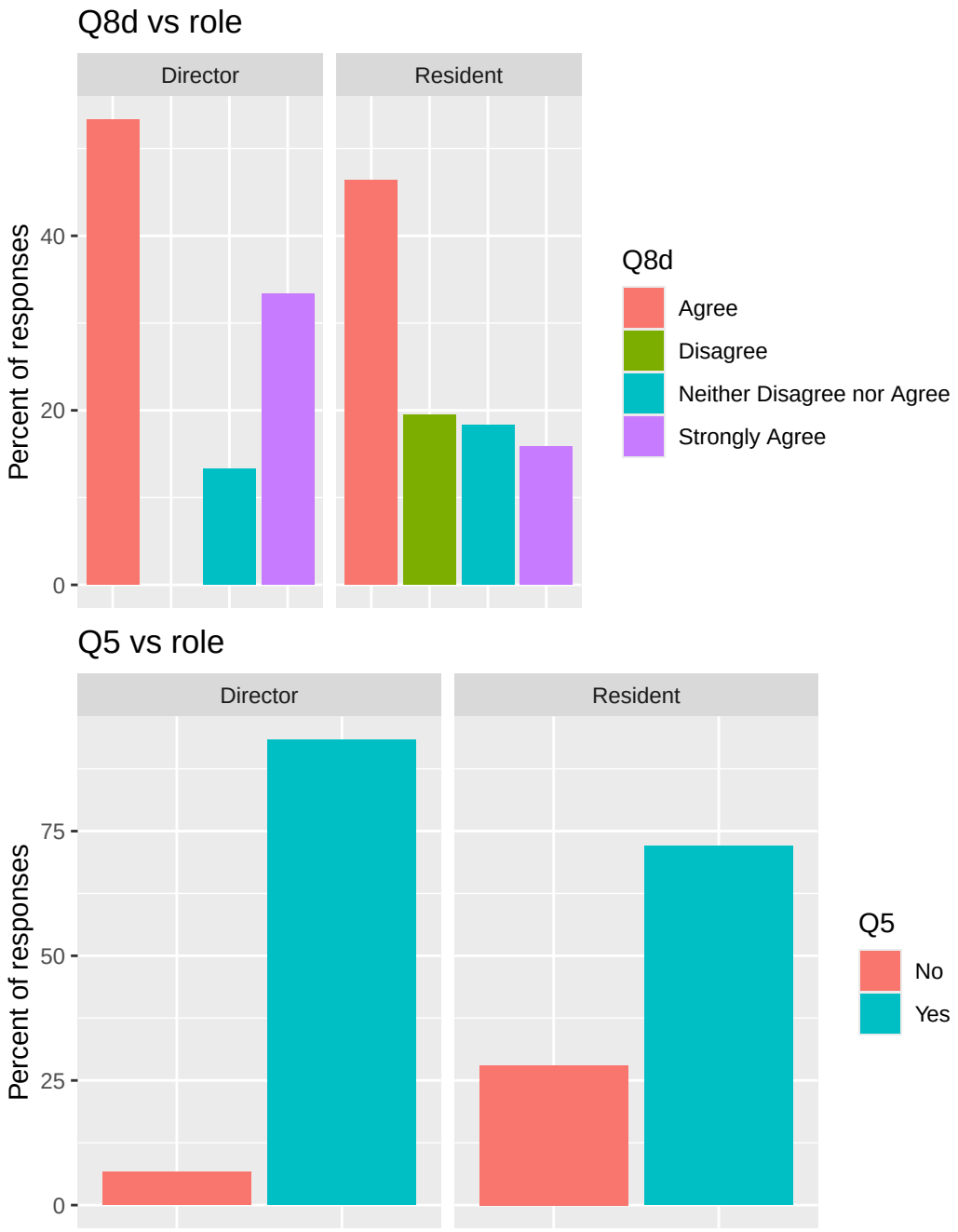

Q3 vs role
Director
Resident
Percent of responses
75
50
25
0
Q3
My program is affiliated with a cleft team or center
No
Unsure
Yes
Q4 vs role
Director
Resident
Percent of responses
75
50
25
0
Q4
My institution does not have a formal cleft team
No
Unsure
Yes

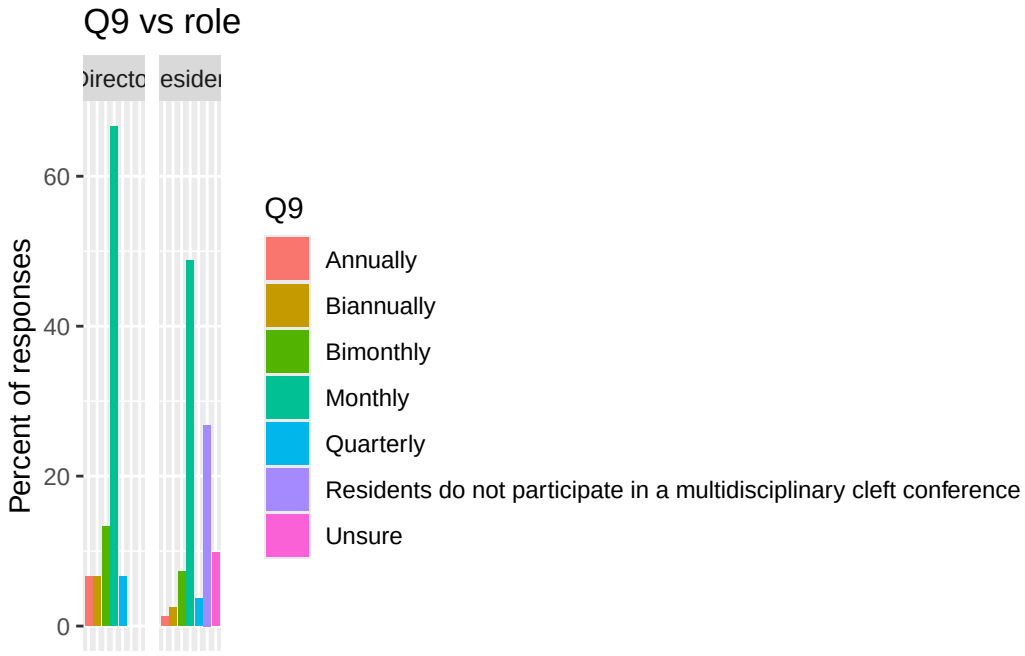

Q6 vs role
Director
Resident
Percent of responses
100
75
50
25
0
Q6
No
Unsure
Yes
Q7a vs role
Director
Resident
Percent of responses
40
30
20
10
0
Q7a
Identify
Manage
Refer

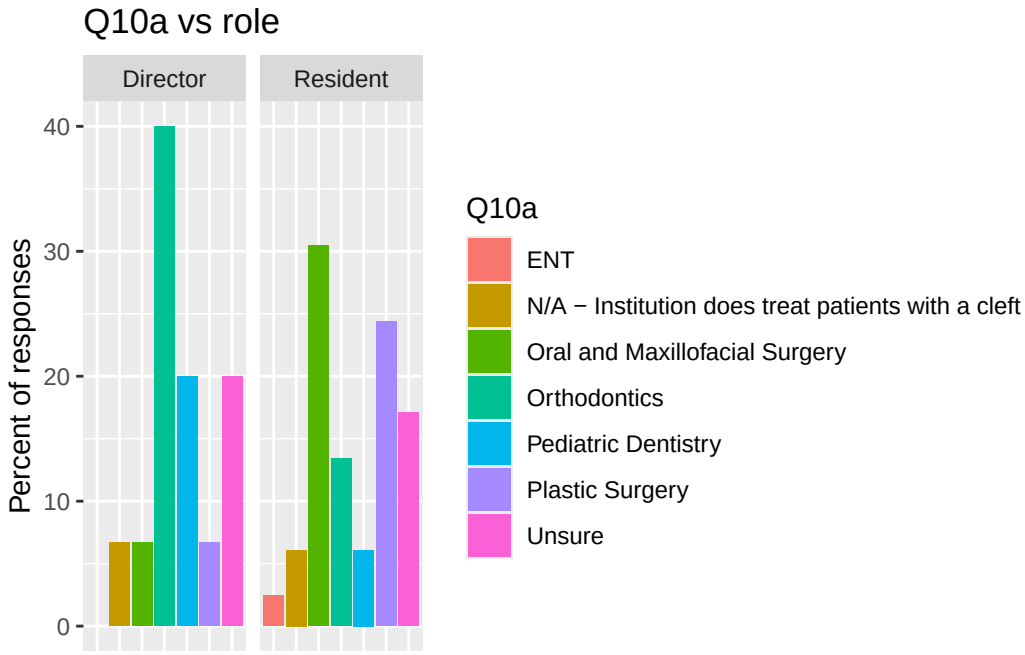

Q7b vs role
Director
Resident
Percent of responses
60
40
20
0
Q7b
Identify
Manage
Refer
Q7c vs role
Director
Resident
Percent of responses
60
40
20
0
Q7c
Identify
Manage
Refer

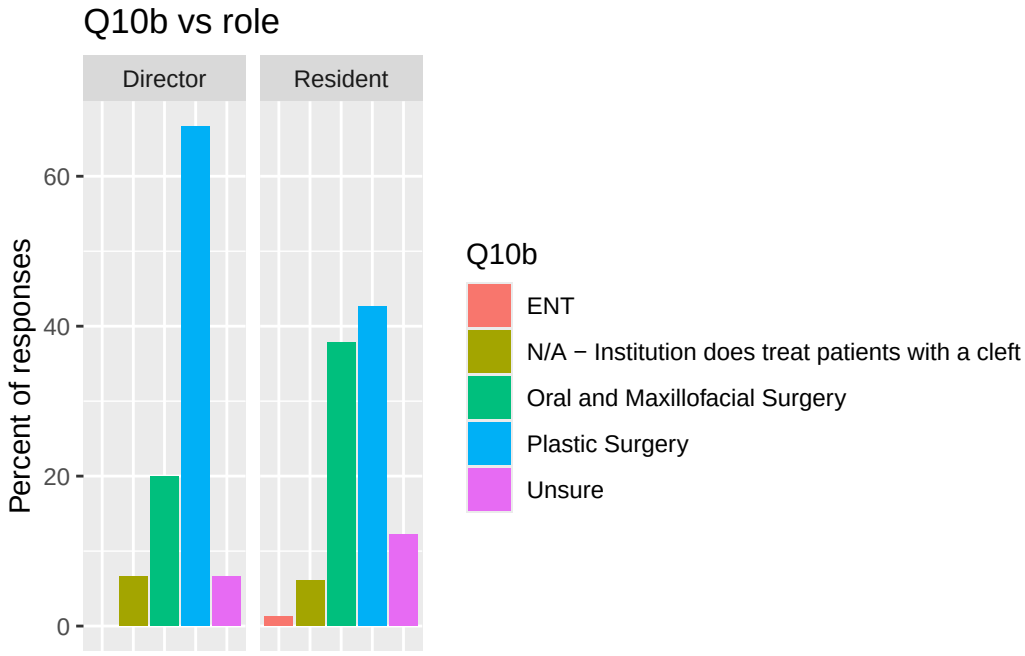

Q7d vs role
Director
Resident
Percent of responses
0
20
40
60
Q7d
Identify
Manage
Refer
Q8a vs role
Director
Resident
Percent of responses
0
20
40
Q8a
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree

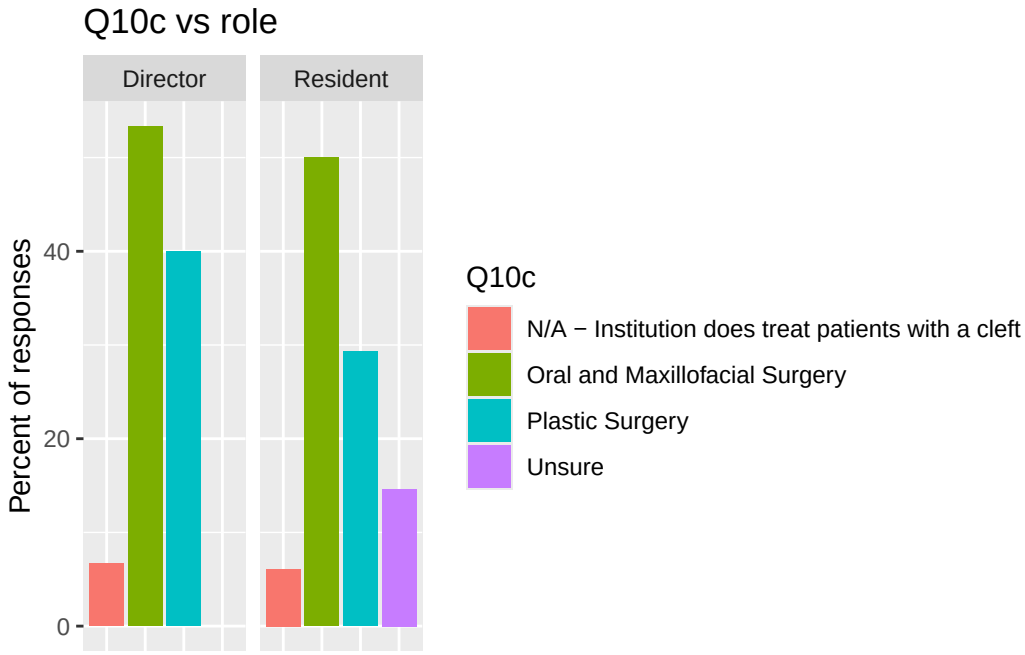

Q8b vs role
Director
Resident
Percent of responses
40
20
0
Q8b
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Q8c vs role
Director
Resident
Percent of responses
40
20
0
Q8c
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree

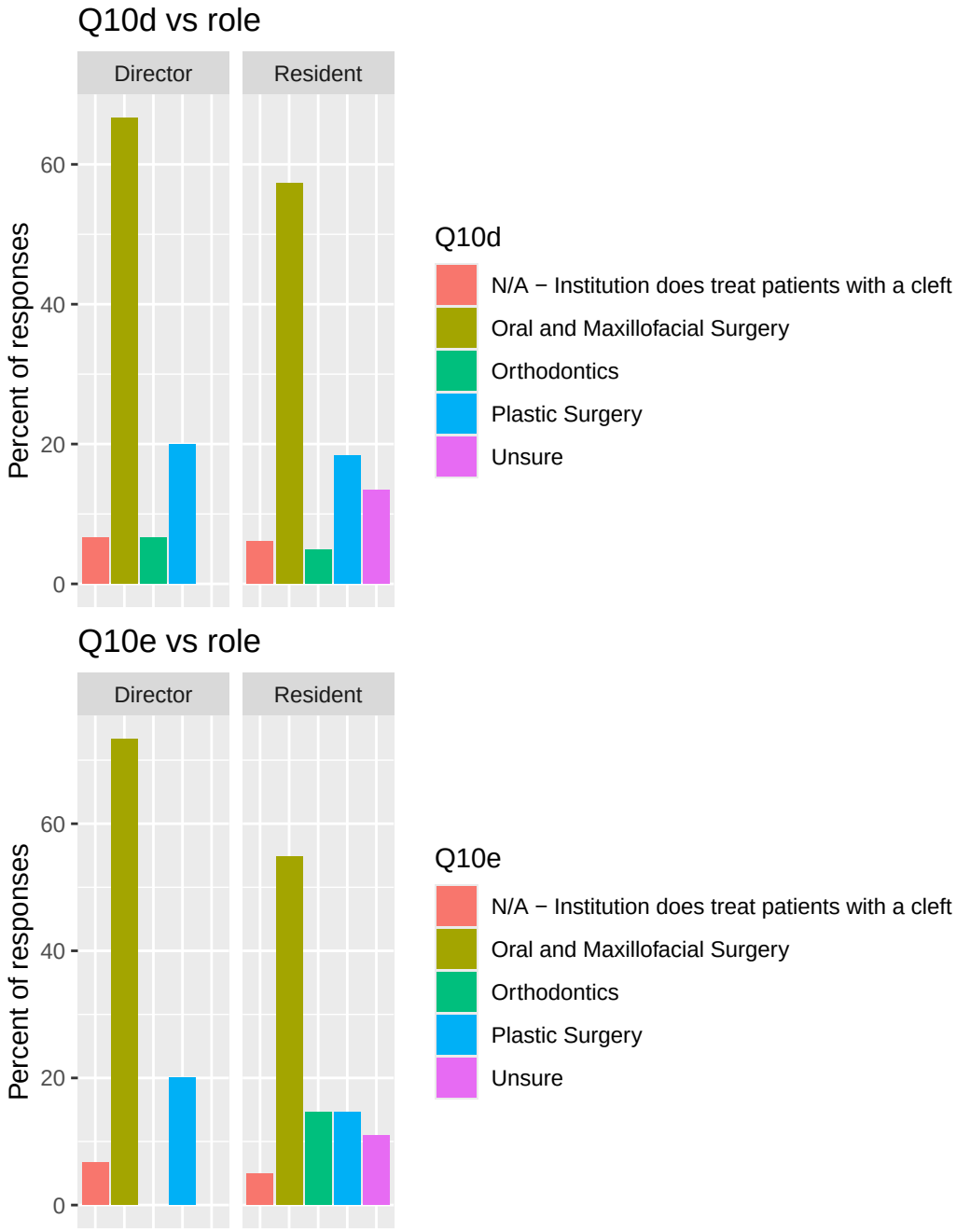

Q8d vs role
Director
Resident
Percent of responses
40
20
0
Q8d
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Q5 vs role
Director
Resident
Percent of responses
75
50
25
0
Q5
No
Yes

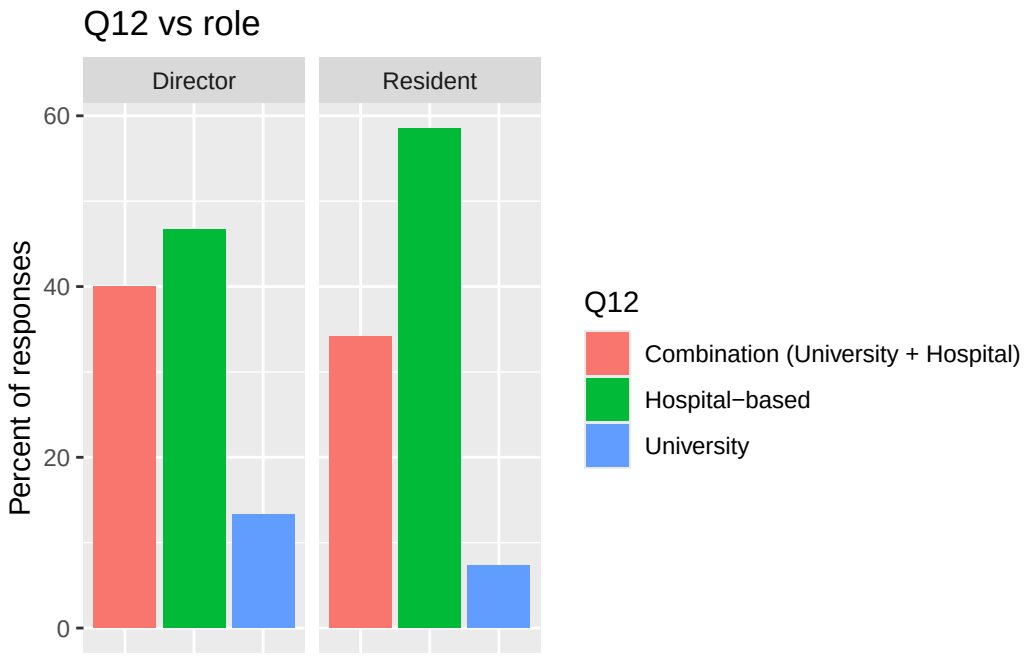

Q9 vs role
Directo
esider
60
40
20
0
Percent of responses
Q9
Annually
Biannually
Bimonthly
Monthly
Quarterly
Residents do not participate in a multidisciplinary cleft conference
Unsure


Q10a vs role
Director
Resident
40
30
20
10
0
Percent of responses
Q10a
ENT
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Orthodontics
Pediatric Dentistry
Plastic Surgery
Unsure


Q10b vs role
Director
Resident
Percent of responses
60
40
20
0
Q10b
ENT
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Plastic Surgery
Unsure


Q10c vs role
Director
Resident
Percent of responses
40
20
0
Q10c
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Plastic Surgery
Unsure


Q10d vs role
Director
Resident
Percent of responses
60
40
20
0
Q10d
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Orthodontics
Plastic Surgery
Unsure
Q10e vs role
Director
Resident
Percent of responses
60
40
20
0
Q10e
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Orthodontics
Plastic Surgery
Unsure



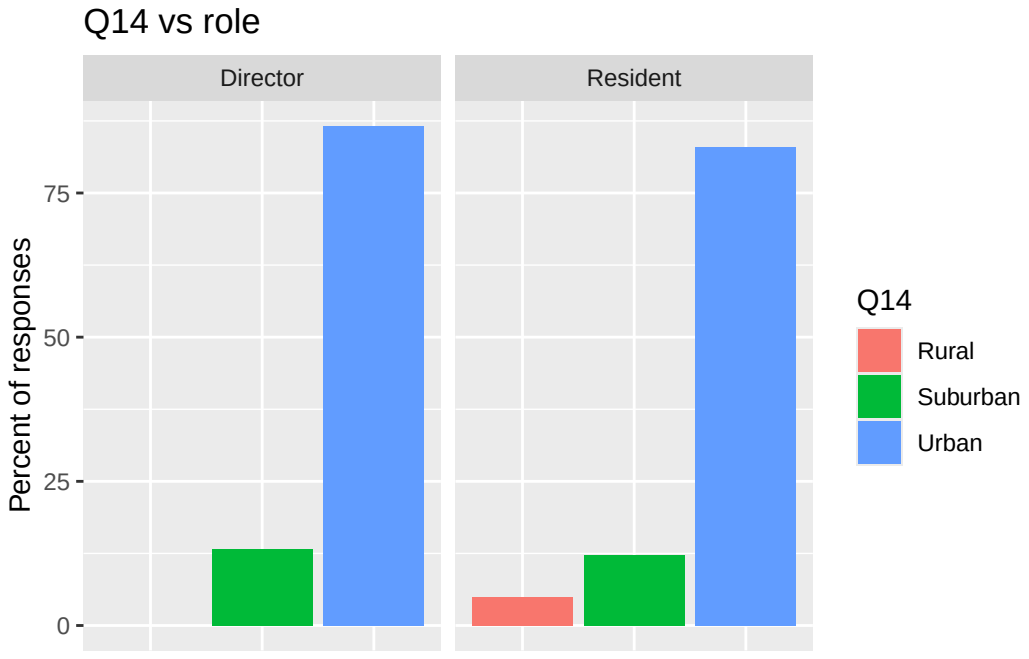
ı, Ohio, South Dakota, Wisconsin, and the Canadian provinces of Manitoba and Ontario)

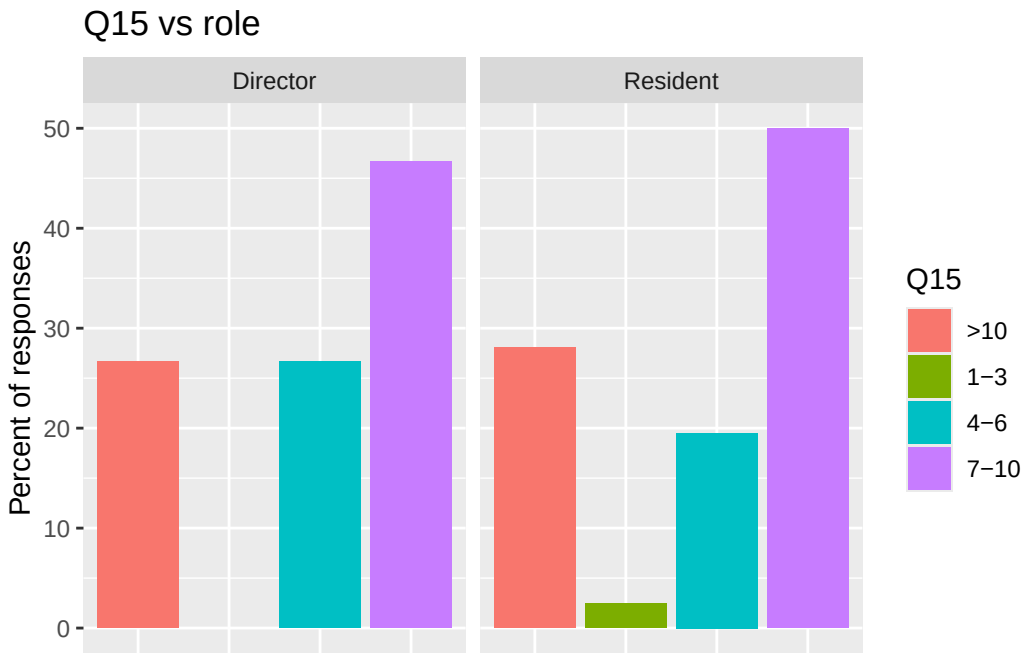
usetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont, and the Cana

ı Carolina, Tennessee, Virginia, West Virginia, the Commonwealth of Puerto Rico, and the US Virgin

and Mexico

Washington, Wyoming, and the Canadian provinces of Saskatchewan, Alberta, British Columbia, No





Appendix B: Code

```
library(tidyverse)
library(zoo)
library(stringr)
library(gt)

i_res_headers <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/CSV/res_survey.csv", n_max

questions <- as.character(i_res_headers[1,])

answer_choices <- as.character(i_res_headers[2,])

questions_filled <- na.locf(ifelse(questions == "" | is.na(questions), NA, questions))

clean_headers <- paste(questions_filled, answer_choices, sep = "_")

i_res_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/CSV/res_survey.csv", skip = 2,
i_res <- i_res_raw %>%
  select(c(10:ncol(i_res_raw)))
```

```r
i_res <- i_res %>%
  mutate(
    across(
      everything(),
      ~ na_if(as.character(.), "")
    )
  )
i_res <- i_res %>%
  select(!"Which departments are primarily responsible for management of the following proce
i_res_q <- i_res %>%
  mutate(Q1 = do.call(coalesce, pick(1:2))) %>%
  mutate(Q2 = do.call(coalesce, pick(3:5))) %>%
  mutate(Q3 = do.call(coalesce, pick(6:9))) %>%
  mutate(Q4 = do.call(coalesce, pick(10:13))) %>%
  mutate(Q5 = do.call(coalesce, pick(14:15))) %>%
  mutate(Q6 = do.call(coalesce, pick(16:18))) %>%
  mutate(Q7a = do.call(coalesce, pick(19:21))) %>%
  mutate(Q7b = do.call(coalesce, pick(22:24))) %>%
  mutate(Q7c = do.call(coalesce, pick(25:27))) %>%
  mutate(Q7d = do.call(coalesce, pick(28:30))) %>%
  mutate(Q8a = do.call(coalesce, pick(31:35))) %>%
  mutate(Q8b = do.call(coalesce, pick(36:40))) %>%
  mutate(Q8c = do.call(coalesce, pick(41:45))) %>%
  mutate(Q8d = do.call(coalesce, pick(46:50))) %>%
  mutate(Q9 = do.call(coalesce, pick(51:57))) %>%
  mutate(Q10a = do.call(coalesce, pick(58:64))) %>%
  mutate(Q10b = do.call(coalesce, pick(65:71))) %>%
  mutate(Q10c = do.call(coalesce, pick(72:78))) %>%
  mutate(Q10d = do.call(coalesce, pick(79:85))) %>%
  mutate(Q10e = do.call(coalesce, pick(86:92))) %>%
  mutate(Q11 = do.call(coalesce, pick(93:94))) %>%
  mutate(Q12 = do.call(coalesce, pick(95:97))) %>%
  mutate(Q13 = do.call(coalesce, pick(98:102))) %>%
  mutate(Q14 = do.call(coalesce, pick(103:105))) %>%
  mutate(Q15 = do.call(coalesce, pick(106:109)))

i_res_q$role <- "Resident"

i_res_coalesced <- i_res_q %>%
  select(110:135)
i_dir_headers <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/i_directors.csv", n_max = 1
```

```r
questions_dir <- as.character(i_dir_headers[1,])

answer_choices_dir <- as.character(i_dir_headers[2,])

questions_filled_dir <- na.locf(ifelse(questions_dir == "" | is.na(questions_dir), NA, questi

clean_headers_dir <- paste(questions_filled_dir, answer_choices_dir, sep = "_")

i_dir_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/i_directors.csv", skip = 2, col
i_dir <- i_dir_raw %>%
  select(c(10:ncol(i_dir_raw)))

i_dir <- i_dir %>%
  mutate(
    across(
      everything(),
      ~ na_if(as.character(.), "")
    )
  )
i_dir <- i_dir %>%
  select(!"Which departments are primarily responsible for management of the following proce
i_dir_q <- i_dir %>%
  mutate(Q1 = do.call(coalesce, pick(1:2))) %>%
  mutate(Q2 = do.call(coalesce, pick(3:5))) %>%
  mutate(Q3 = do.call(coalesce, pick(6:9))) %>%
  mutate(Q4 = do.call(coalesce, pick(10:13))) %>%
  # mutate(Q5 = do.call(coalesce, pick(14:25))) %>%
  mutate(Q6 = do.call(coalesce, pick(26:30))) %>%
  # mutate(Q7a = do.call(coalesce, pick(31:37))) %>%
  mutate(Q8 = do.call(coalesce, pick(38:42))) %>%
  mutate(Q9 = do.call(coalesce, pick(43:45))) %>%
  mutate(Q10a = do.call(coalesce, pick(46:48))) %>%
  mutate(Q10b = do.call(coalesce, pick(49:51))) %>%
  mutate(Q10c = do.call(coalesce, pick(52:54))) %>%
  mutate(Q10d = do.call(coalesce, pick(55:57))) %>%
  mutate(Q11a = do.call(coalesce, pick(58:62))) %>%
  mutate(Q11b = do.call(coalesce, pick(63:67))) %>%
  mutate(Q11c = do.call(coalesce, pick(68:72))) %>%
  mutate(Q11d = do.call(coalesce, pick(73:77))) %>%
  mutate(Q12 = do.call(coalesce, pick(78:79))) %>%
  mutate(Q13 = do.call(coalesce, pick(80:86))) %>%
  mutate(Q14 = do.call(coalesce, pick(87:90))) %>%
```

```
  mutate(Q15 = do.call(coalesce, pick(91:94))) %>%
  mutate(Q16a = do.call(coalesce, pick(95:101))) %>%
  mutate(Q16b = do.call(coalesce, pick(102:108))) %>%
  mutate(Q16c = do.call(coalesce, pick(109:115))) %>%
  mutate(Q16d = do.call(coalesce, pick(116:122))) %>%
  mutate(Q16e = do.call(coalesce, pick(123:129))) %>%
  mutate(Q17 = do.call(coalesce, pick(130:131))) %>%
  mutate(Q18 = do.call(coalesce, pick(132:134))) %>%
  mutate(Q19 = do.call(coalesce, pick(135:139))) %>%
  mutate(Q20 = do.call(coalesce, pick(140:142))) %>%
  mutate(Q21 = do.call(coalesce, pick(143:146)))

i_dir_coalesced_partial <- i_dir_q %>%
  select(147:150, 14:25, 151, 31:37, 152:175)

i_dir_coalesced <- i_dir_coalesced_partial %>%
  mutate(Q5 = pmap_chr(select(., 5:16), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q7 = pmap_chr(select(., 18:24), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  select(1:4, 49, 17, 50, 25:48)

i_dir_coalesced$role <- "Director"
i_dir_coalesced <- i_dir_coalesced %>%
  mutate(Q21_new = case_when(
    Q21 == "3-Jan" ~ "1-3",
    Q21 == "6-Apr" ~ "4-6",
    Q21 == "10-Jul" ~ "7-10",
    Q21 == ">10" ~ ">10"
  ))
# 2 with 7 and 8
# 2 with 11
# 2 with 11 and 12
question1s <- c("Q7a", "Q7b", "Q7c", "Q7d", "Q8a", "Q8b", "Q8c", "Q8d", "Q11", "Q12")

question2s <- c(rep("Q2", 10))

question_dic_res <- data.frame(question1s, question2s)
validation <- function(question1, question2, data){
  df <- data %>%
    transmute(
      response1 = .data[[question1]],
      response2 = .data[[question2]]
    ) %>%
```

```
    drop_na()

  comparison_res <- df %>%
    group_by(response1, response2) %>%
    summarize(counts = n(), .groups = "drop") %>%
    group_by(response2) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  table_comparison <- table(df$response2, df$response1)
  #table_hours
  test <- suppressWarnings(chisq.test(table_comparison))

  if (mean(test$expected <= 5) < 0.8 | any(test$expected < 1)){
    fish_test <- fisher.test(table_comparison, simulate.p.value = TRUE)
    if (fish_test$p.value <= 0.05){
      significant <- TRUE
    }
    else {
      significant <- FALSE
    }
    return(c("question1", "question2", round(fish_test$p.value, digits = 4), significant, "fi
  }
  else{
    if (test$p.value <= 0.05){
      significant <- TRUE
    }
    else {
      significant <- FALSE
    }
  }
  return(c("question1", "question2", round(test$p.value, digits = 4), significant, "chisq"))
}
res_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(question_dic_res))) {
```

```
  res_results[i, ] <- validation(
    question1 = question_dic_res$question1s[i],
    question2 = question_dic_res$question2s[i],
    data = i_res_coalesced
  )
}

res_results <- res_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = question_dic_res$question1s,
    Question2 = question_dic_res$question2s
  ) %>%
  arrange(Adj_P)

gt(res_results)
library(modelsummary)
cleft_program <- i_res_coalesced %>%
  filter(Q2 != "Unsure") %>%
  mutate(cleft_response = as.factor(Q11),
         program = as.factor(Q12),
         cleft_center = as.factor(Q2)) %>%
  select(cleft_response, cleft_center, program) %>%
  drop_na()
cleft_glm <- glm(cleft_response ~ cleft_center + program, family = "binomial", data = cleft_p
cleft_null <- glm(cleft_response ~ 1, family = "binomial", data = cleft_program)
anova(cleft_glm, cleft_null)

#logistic regression
#use model summary
# we notice signficance when comparing the model with program and cleft centers to the null m
modelsummary(cleft_glm)
common_qs_res <- c("Q2", "Q3", "Q4", "Q6", "Q7a", "Q7b", "Q7c", "Q7d","Q8a","Q8b", "Q8c", "Q8
                   "role")
common_qs_dir <- c("Q2", "Q3", "Q4", "Q9", "Q10a", "Q10b", "Q10c", "Q10d", "Q11a","Q11b","Q11
                  "Q21_new",   "role")
q_dic <- data.frame(common_qs_res, common_qs_dir)

res_common <- i_res_coalesced %>%
  select(all_of(common_qs_res))
```

```r
dir_common <- i_dir_coalesced %>%
  select(all_of(common_qs_dir)) %>%
  setNames(common_qs_res)
res_dir <- bind_rows(
  res_common,
  dir_common
)
res_dir_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(ncol(res_dir)-1)) {

  res_dir_results[i, ] <- validation(
    question1 = colnames(res_dir)[i],
    question2 = colnames(res_dir)[ncol(res_dir)],
    data = res_dir
  )
}

res_dir_results <- res_dir_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = colnames(res_dir)[1:(ncol(res_dir) -1)],
    Question2 = "role"
  ) %>% arrange(Adj_P)
gt(res_dir_results)
stack_bar <- function(q_reference, q_group, data){
  bar_df <- data %>%
    group_by(.data[[q_reference]], .data[[q_group]]) %>%
    summarize(counts = n(), .groups = "drop") %>%
    drop_na() %>%
    group_by(.data[[q_group]]) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  p <- ggplot(bar_df, aes(x = .data[[q_reference]], y = percent, fill = .data[[q_reference]]
```

```
    geom_col() +
    facet_wrap(vars(.data[[q_group]])) +
    labs(
      title = paste(q_reference, "vs", q_group, sep = " "),
      x = NULL,
      y = "Percent of responses"
    ) +
    theme(
      axis.text.x = element_blank(),
      axis.ticks.x = element_blank()
    )
  return(p)
}
plots_res <- pmap(question_dic_res,
                  function(question1s, question2s){
                    stack_bar(
                      q_reference = question1s,
                      q_group = question2s,
                      data = i_res_coalesced
                    )
                  })
walk(plots_res, print)

question_dic_res_dir <- data.frame(question1s_res_dir = colnames(res_dir)
                                   , question2s_res_dir = rep("role", ncol(res_dir)))

question_dic_res_dir <- question_dic_res_dir %>%
  filter(question1s_res_dir != "role")
plots_comp <- pmap(question_dic_res_dir,
                   function(question1s_res_dir, question2s_res_dir){
                     stack_bar(
                       q_reference = question1s_res_dir,
                       q_group = question2s_res_dir,
                       data = res_dir
                     )
                   })
walk(plots_comp, print)
```